

GARRETT COMES

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SIMULATION & SCIENTIFIC SOFTWARE

Valurile — GPU Lattice-Boltzmann CFD Solver | Rust, CUDA

2024 – present

- Multi-crate LBM solver built for memory-bandwidth-bound performance, with Esoteric Pull in-place streaming that roughly halves memory traffic per timestep, static mesh refinement, Smagorinsky–Lilly LES, and SRT/TRT/MRT collision operators.
- Interactive egui/GPU analysis frontend with airfoil and cylinder obstacles, physical-to-lattice nondimensionalization, velocity/density/pressure/vorticity fields, and drag and lift coefficient histories.

YBCO Coil Quench Simulation — SCRAP senior design | Python, FEniCSx, Gmsh, PETSc

2025 – 2026

- Coupled electromagnetic–thermal model of a high-temperature superconducting pancake coil, with H-formulation EM using E–J power-law resistivity, per-turn tape meshing, staggered thermal coupling with radiation and cold-finger BCs, quench detection, and adaptive time-stepping, supporting an AIAA SciTech 2027 extended abstract in preparation.

spacedb — Space-Object Database & Route Planner | Rust, egui, wgpu

2025 – 2026

- Native GUI and CLI over the full public satellite and debris catalog, with SGP4 and high-fidelity propagation under drag, SRP, and third-body effects, parallel collision screening, decay prediction, value scoring, and multi-target servicing-route optimization in an interactive wgpu 3D scene.

ai-cfd — GPU-Resident Solver for ML Dataset Generation | Python, Triton

2026

- Staggered-grid incompressible flow solver written entirely in Triton kernels, with adaptive timestep, Jacobi/CG/BiCGStab pressure solves, per-step divergence, residual, and drag/lift diagnostics, and traceable train/val/test dataset sweeps for AI-CFD surrogates.

fem.rust.ae — FEM Solver in Rust, live in the browser | Rust, WASM, Bevy

2025

- From-scratch structural FEM with sparse solves via faer, an interactive 3D UI, and mesh-refinement convergence studies, compiled to WebAssembly and running live at garrettcomes.me/fem.rust.ae.

Thermodynamic Sampling Research | JAX, THRML, PyTorch

2026

- Authored a preprint on when thermodynamic sampling hardware (Extropic's TSU) beats GPUs on the energy cost of protein and drug design, about $10^6\times$ less sampling energy for equal library coverage across 59 Potts landscapes, with the end-to-end ratio honestly bounded by digital readout and matching. Earlier work hybridized a masked-diffusion LM with THRML joint decoding, beating independent argmax by 7–15 pp, reproducible bit-for-bit with 9/9 regression tests.

Eddy-Current Braking System — Controls & Hardware | Rust no_std, Arduino Due, NGSolve

2025

- End-to-end controls project with no_std Rust firmware, magnetic encoder feedback, and feedforward-plus-PI speed control, a live egui tuning dashboard over UART telemetry plotting internal controller signals, and Python/NGSolve magnetic simulations of the braking physics.

EXPERIENCE

Lockheed Martin Space — Propulsion Engineering Intern, Civil & Commercial Space

Summers 2022 – 2025

Flight programs incl. Dragonfly and Mars Sample Return

Waterton Campus, Denver, CO

- Designed a piston-tank propellant gauging system in which hermetically sealed internal magnets are read by rad-hard Hall-effect sensors, sized with optimization-driven magnetic simulation. Built propellant slosh and FSI models in Ansys and pure slosh CFD in FLOW-3D, correlated against a test rig, and contributed to a hydrazine-replacement green-propellant trade study.

EDUCATION

Georgia Institute of Technology

Jul 2022 – Aug 2026

B.S. Aerospace Engineering, Concentration in Propulsion

Atlanta, GA

- Coursework: Computational Fluid Dynamics, Finite Element Analysis, Jet & Rocket Propulsion, Aerothermodynamics, Control System Design.

SKILLS

Core: Rust, C++, Python, CUDA/Triton, JAX, PyTorch, FEM/LBM/FVM numerical methods, GPU memory-bandwidth optimization, wgpu/OpenGL

Tooling: Linux/NixOS, Git, MPI/PETSc/FEniCSx, eBPF, WASM, agentic AI development with Claude Code, MCP servers, and custom agents